

Original Research Article

Low clarithromycin resistance in virulent *Helicobacter pylori* from dyspeptic patients at a tertiary care centre in Odisha

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ABSTRACT

Purpose: Universal eradication or use of failing antibiotic can add fuel to the antimicrobial resistance pandemic. Outcome of *Helicobacter pylori* (HP) infection depends at least partly virulence factors and its eradication as preventive measure against gastric cancer is advocated by some guidelines. There is need to identify candidates at risk for gastric cancer and antimicrobial resistance in HP for rational management. Such candidates could be identified by studying the association of virulence factors with clinical outcome. As this data is lacking from Odisha this study was undertaken.

Methods: 113 consecutive dyspeptic patients who underwent endoscopy at our hospital were recruited to obtain gastric biopsies for culture and antibiotic susceptibility, histological examination, molecular detection of HP, virulence typing (*cagA*, EPIYA typing, *vacA*, *vacA s1/s2*, *vacA m1/m2* and *babA2*) by conventional PCR and identification of clarithromycin resistance by real-time PCR. Cultured isolates were subjected to antibiotic sensitivity using e strips as per EUCAST guidelines.

Results: 93 (82.3%) dyspeptic patients were infected by HP by histology & PCR, while 90 (79.6%) were rapid urea test positive, and HP was cultured from 32 (28.3%) of these patients. Eleven (11.8%) of the 93 samples with HP were resistant to clarithromycin by real-time PCR. Of the 93 patients with HP infection by histopathology and PCR, 62 (66.7%), 87 (93.5%) and 43 (46.2%) harboured *cagA*, *vacA* and *babA2* genes. The western *cagA* found in 33 (35.5%) samples and *vacA s1m1* in 50 (53.8%) samples were the commonest virulence subtypes. No association was found between virulence factors and outcome except *vacA s2m2* and *vac s1/m1m2*, which were significantly associated with peptic ulcers. Phenotypically 11 (34.4%), 1 (3.1%), 21 (65.6%) and 26 (81.2%) isolates were resistant to clarithromycin, amoxicillin, levofloxacin, and metronidazole.

Conclusions: This is the first study that explored the antibiotic resistance of HP, and its virulence factors in dyspeptic patients from this region of India.

1. Introduction

Helicobacter pylori (HP) is a curved gram-negative urease-producing bacterium that can be found in the mucus coating the gastric mucosa or between the mucous layer and gastric epithelium. It is estimated to infect around half of the world population in whom it causes chronic gastritis [1]. Untreated chronic gastritis can lead to the development of a variety of gastroduodenal disorders ranging from peptic ulcer disease to gastric cancer. The outcome of the infection at least partly depends on the

presence of bacterial virulence factors like *cagA*, *vacA*, *babA*, *dupA*, *iceA* etc [2]. While, studies from the west have found an association of virulence factors with the outcome; studies from East Asia have not corroborated with the above pointing towards geographic variation [3]. Universal eradication regimens advocated in some guidelines can add fuel to the antimicrobial pandemic [4,5]. Hence, there is a need to identify candidates colonized by virulent HP and at risk for gastric cancer eradication therapy. If an association is found between HP virulence factors with clinical outcome this information could be used to identify

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patients for eradication therapy. Further, a knowledge of resistance to clarithromycin which is used in first line, quadruple, sequential, concomitant, and hybrid therapies for HP eradication is essential for patient management [6]. Because data on the virulence types, their association with outcome and antibiotic susceptibility of HP from Odisha India are not available, we undertook this study in patients with dyspepsia attending our hospital.

2. Materials and methods

2.1. Patients recruitment & gastric biopsy sampling

Consecutive dyspeptic patients with alarm features undergoing upper gastrointestinal tract endoscopy and consented for study at the department of gastroenterology, All India Institute of Medical Sciences, Bhubaneswar between October 2018 to March 2020 were recruited for the study. Patients with active bleeding or history of intake of antibiotics in last month or proton pump inhibitors within the previous two weeks were excluded from the study. A total of 4 biopsies were obtained 2 each from antrum & corpus. Biopsies were sent in formol saline for histopathology and stuart's urease test broth with 20% glucose to microbiology laboratory within 1 h of collection.

2.2. Histopathology

Biopsies were reported as per the Sydney classification for chronic gastritis [7]. Histopathology was considered as reference for diagnosis of HP infection.

2.3. Molecular detection of *Helicobacter pylori* and virulence genes

Biopsies obtained from antrum and body were pooled and DNA was extracted using the QIAamp DNA Mini Kit (Qiagen, Valencia, CA, USA) as per the manufacturer's instructions. Polymerase Chain Reaction (PCR) was performed to detect the presence of HP by using primers for the *glmM* gene [8]. Samples which tested positive for *glmM* gene were subjected to a series of multiplex and simple PCR to detect the presence of *cagA*, *vacA* subtype (*s1/s2* & *m1/m2*) and *babA2* as described earlier [9–11]. The details of the primers used in this study are given in Table 1. All reactions were performed using the KAPA2G Multiplex PCR Master Mix (Sigma Aldrich, India). The master-mix for each reaction contained the final reaction volume of 25 µl with 12.5 µl of 2X Multiplex PCR Master Mix, primers at a final concentration of 0.3 µM and 5 µl of template DNA. The PCR products were electrophoresed on 2% agarose gel and after ethidium bromide staining the DNA bands visualized under a UV transilluminator and photographed. Each batch of reactions was accompanied by DNA

extracted from *Helicobacter pylori* ATCC 26695 as positive control and autoclaved sterile double distilled water as no template control.

2.4. Realtime PCR for clarithromycin resistance

Clarithromycin resistance was determined using the VIASURE *H. pylori* + Clarithromycin resistance Real-Time PCR Detection Kit (VS-CLA106L). All samples which tested positive for HP by inhouse *glmM* PCR were tested for clarithromycin resistance as per manufacturer's instructions.

2.5. Culture & phenotypic tests

The results of the rapid urea test was noted from the change in color of the transport medium from yellow orange to pink color within 30 min of placing the biopsies into it. In the laboratory biopsies were retrieved from the transport medium and ground together in brain heart infusion broth using a micropestle. Thirty microlitre each of the homogenate was cultured on 5% commercial Columbia sheep blood agar (Himedia) and Columbia agar with dent supplement and 7% laked human blood. It was then incubated in BBL™ CampyPak™ Plus at 37°C and checked on day 3 and day 7 for suspicious colonies. Translucent dew drop like colonies were presumptively identified as *H. pylori* if found to produce oxidase and rapidly break down urea on Christensen's urea agar. These were confirmed by *glmM* PCR as detailed above. Susceptibility was performed for Clarithromycin, Amoxicillin, Levofloxacin, and metronidazole using Himedia EZY MIC strips as per British Society of Antimicrobial chemotherapy (BSAC) guidance using and interpreted as per BSAC guidelines which are similar to EUCAST guidelines [12].

2.6. Statistical analyses

Data was entered in Microsoft excel and then imported to SPSS 23 for statistical analysis. Comparison between groups were made using the students-t test, Chi-square test and Fisher-exact test as applicable. p values < 0.05 were considered significant.

2.7. Ethics

The study was performed after obtaining ethics approval from the institute ethics committee. The approval number of the study was T/IM-F/17–18/10. Written informed consent was obtained from each patient enrolled in the study.

Table 1
PCR primers for the amplification of *H. pylori* virulence genes.

Gene amplified	Primer name	Primer sequence (5' → 3')	Product Size (bp)
<i>glmM</i> [8]	GlmM –F GlmM –R	GGATAAGCTTTTAGGGGTGTTAGGGG GCTTACTTTCTAACACTAACGCGC	296
<i>vacA</i> s1/ <i>vacA</i> s2 [9]	VAI-F VAI-R	ATGGAAATACAACAACACAC CTGCTTGAATGCGCCAAAC	s1, 259 s2, 286
<i>vacA</i> m1/ <i>vacA</i> m2 [9]	VAG-F VAG-R	CAATCTGTCCAATCAAGCGAG GCGTCAAAATAATTCCAAGG	m1, 567 m2, 642
<i>cagA</i> [9]	cag5c-F cag3c-R	GTTGATAACGCTGTCGCTTC GGGTTGTATGATATTTCCATAA	350
Forward primer for all EPIYA motif of <i>cagA</i> [11]	cagA28F	TTCTCAAAGGAGCAATTGGC	
EPIYA-A [11]	cagA-P1C	GTCTGCTTTCTTTTATTAACCTKAGC	264
EPIYA-B [11]	cagA-P2TA	TTTAGCAACTTGAGTATAAATGGG	306
EPIYA-C [11]	cagAWest	TTTCAAAGGGAAAGGTCGCGC	501
EPIYA-D [11]	cagAEast	AGAGGGAAGCCTGCTTGATT	495
<i>babA2</i> [10]	nthu_babA2F babA2R babA2F babA2R607	AATCGAAAAAGGAGAAAACATGAAAAA TGTTAGTGATTTCGGTGTAGGACA AATCCAAAAAGGAGAAAAGTATGAAA GTTTCTTTGAGCGCGGGTAAGC	832 607 480

3. Results

3.1. Demographic details

A total of 85 males and 28 females were enrolled in the study. Of these 113 patients, 63 (55.8%) had non-ulcer dyspepsia, 35 (31%) had peptic ulcer disease, and the remaining 15 (13.3%) had gastric carcinoma.

3.2. Histopathology

HP was found in 93 (82.3%) on histological examination. The absence of inflammation and neutrophilic activity were significantly associated with the lack of HP infection. At the same time, mild neutrophilic activity was significantly associated with HP infection (Table 2). Because of low numbers, cases with gastric cancer were not included in the analysis.

3.3. Molecular detection of *H. pylori*, virulence genes and their association with clinical outcome

Specific PCR products using primers mentioned in the Table 1 were visualised after gel electrophoresis (see Fig. 1). Patients 93 (82.3%) who had HP on histopathology were also positive by *glmM* PCR, and no false positives were observed by *glmM* PCR. The mean age of patients infected by HP was 46.85 ± 13.2 . A high prevalence of HP was found in all gastroduodenal diseases, with the highest in patients with peptic ulcer (31/35, 88.6%), followed by patients with non-ulcer dyspepsia (53/63, 84.1%) and gastric carcinoma (9/15, 60%). *CagA*, *vacA* and *babA2* genes were present in 62(66.7%), 87(93.5%) and 43 (46.2%) samples with HP.

The distribution of various virulence factors in different clinical conditions are given in Table 3. EPIYA motifs could be amplified in 53 of the 62 *cagA* positive samples. EPIYA ABC, BC, AC and AB were found in 33(53.2%), 15(24.2%), 4(6.5%) and 1 (1.6%) strains of the 62 *cagA* positive samples. *VacA s1*, *VacA s2*, *VacA m1*, *VacA m2* were present in 77(82.8%), 10(10.8%), 55(59.1%), and 9 (9.7%) of HP containing samples. Among the different combinations of *vacAs1/s2* and *vacAm1/m2*; *vacA s1m1* type in 50 (53.8%) samples was the most prevalent, followed by *vacA s1/m1m2*, which was found in 19 (20.4%) of samples. Rest of the combinations were present in less than 7% of samples containing HP. No statistically significant association was found between the presence of virulence factors and clinical outcome except *vacA s2m2* and *vacA s1/m1m2*, which were significantly associated with peptic ulcers.

3.4. Realtime PCR for clarithromycin resistance

Eleven (11.8%) of the ninety-three samples containing HP by PCR were clarithromycin resistant. Ten of these samples showed the presence of susceptible and resistant populations, while one sample showed the presence of clarithromycin resistance only.

3.5. Culture and phenotypic tests

RUT was positive in 96.7% (90/93 cases), and HP was cultured from 34.4% (32/93) of patients with HP on histopathological examination. On MIC testing, 11(34.4%), 1(3.1%), 21(65.6%) and 26 (81.2%) isolates were resistant to clarithromycin, amoxicillin, levofloxacin, and metronidazole. Details of the susceptibility are given in Table 4.

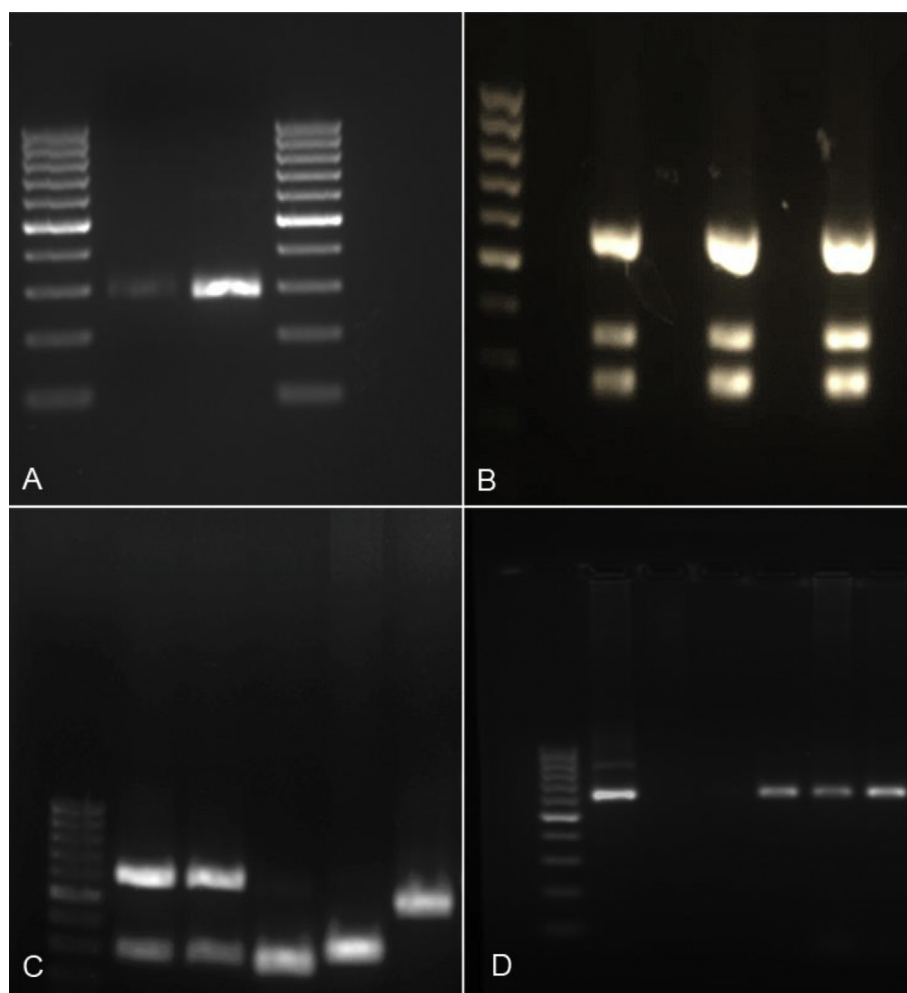


Fig. 1. Representative gel images with lanes read from left to right containing 100 bp ladder. A) *glmM* PCR gel showing 296 bp products in lane 2 and 3 besides DNA ladder in lanes 1 and 4. B) Multiplex PCR (*cagA*, *vacA s1/s2*, and *vacA m1/m2*) gel showing 3 bands of 259 bp, 350 bp and 567 bp corresponding to *vacAs1*, *cagA*, *vacA m1* in lanes 3, 5 and 7 with lane 1 showing DNA ladder and lanes 3, 4 and 5 showing EPIYA A (~264bp), EPIYA B (~306 bp) and EPIYA C (~501bp). D) *babA2* PCR with lanes 1 containing DNA ladder and lanes 4–6 containing a 601 bp product.

Table 2

Showing the association of HP with demographic & histopathological parameters.

	HP positive	HP negative	P value
Age	46.85 ± 13.2	49.7 ± 9.782	0.045
Gender			
Male	73	12	0.07
Female	20	8	
Inflammation			
None	14	11	0.00010*
Mild	30	5	>0.05
Moderate	40	4	>0.05
Severe	9	0	>0.05
Activity			
Absent	34	19	.00000*
Mild	40	1	.00133*
Moderate	14	0	>0.05
Severe	5	0	>0.05
Metaplasia			
Absent	83	18	>0.05
Present	10	2	
Atrophy			
Absent	84	20	>0.05
Mild	6	0	
Marked	3	0	
Malignancy			
Absent	85*	13	0.001
Present	8	7*	0.001

* indicates p values < 0.05

Table 3

Distribution of various virulence factors, showing lack of association of virulence with clinical outcome. Cases with gastric cancer were excluded from analysis due to small number of cases.

Virulence factors	Diagnosis			P value
	Non Ulcer	Peptic Ulcer	Gastric Cancer	
<i>cagA</i>		(19, 61.3%)	(5, 55.6%)	>0.05
	Dyspepsia (38, 66.7%)			
• ABC	25 (65.8%)	7 (36.8%)	1 (20%)	>0.05
• AB	0 (0%)	1 (5.3%)	0 (0%)	>0.05
• AC	3 (7.9%)	1 (5.3%)	0 (0%)	>0.05
• BC	5 (13.2%)	6 (31.6%)	4 (80%)	>0.05
• Negative	5 (13.2%)	4 (21.1%)	0 (0%)	>0.05
<i>vacA</i>		Peptic Ulcer	Gastric Cancer	>0.05
	Non Ulcer	(30, 96.8%)	(7, 77.8%)	
	Dyspepsia (50, 94.3%)			
• <i>s1m1</i>	31 (62%)	12 (40%)	7 (100%)	>0.05
• <i>s1m2</i>	5 (10%)	1 (3.3%)	0 (0.0%)	>0.05
• <i>s1/m1m2</i>	8 (16%)	11 (36.7%)*	0 (0.0%)	0.02
• <i>s1</i>	2 (4%)	0 (0.0%)	0 (0.0%)	>0.05
• <i>s2m1</i>	4 (8%)	1 (3.3%)	0 (0.0%)	>0.05
• <i>s2/m1m2</i>	0 (0.0%)	2 (6.7%)	0 (0.0%)	>0.05
• <i>s2m2</i>	0 (0.0%)	3 (10%)*	0 (0.0%)	0.02
<i>babA2</i>		Peptic Ulcer	Gastric Cancer	>0.05
	Non Ulcer	(12, 38.7)	(3, 33.3%)	
	Dyspepsia (28, 52.8%)			

Table 4

Details susceptibility of *H. pylori* to various drugs tested in this study.

Antibiotic	Resistance in percentage	MIC Range
Clarithromycin	34.4% (11/32)	0.016–2 µg/mL
Amoxicillin	3.1% (1/32)	0.016–0.19 µg/mL
Levofloxacin	65.6% (21/32)	0.032–16 µg/mL
Metronidazole	81.2% (26/32)	4–128 µg/mL

4. Discussion

Helicobacter pylori is only bacterial carcinogen to be listed among the IARC group 1 agent. Around 50% of the world's population is said to

infected by this bacterium in whom it always causes chronic gastritis [1]. Some of these infected individuals then go on to develop varied diseases like peptic ulcer and gastric cancer [1]. The prevalence of HP, its virulence traits, antimicrobial resistance and outcome following infection vary depending on the geographical location [2]. Knowledge of the above factors helps in understanding the pathogenesis and have the potential to alter the management of the disease. For example, in localities with a high prevalence of HP, a test and treat strategy is used for the management of infections and in areas with clarithromycin resistance ≥15% PPI-clarithromycin-containing triple therapy without prior testing should be abandoned [1].

In the past, it was found that the prevalence of HP at the community level ranges from 70 to 90% in developing countries and between 25 and 50% in developed countries [13]. In a recent meta-analysis, the prevalence of HP in India in the community was estimated to be 63.5% [14]. This hospital-based study from eastern India found a higher positivity of 82.3% HP infection in our study population. Although, majority of the studies from India, including those from Odisha, have reported positivity of HP between 30 and 60% and only a few studies have reported a prevalence of more than 70% [15]. Such a wide variation could be because of differences in levels of sanitation, access to safe drinking water, socioeconomic status across the country, or due to referral bias, and the study does not reflect HP infection in the community [16].

Approximately 66.7% of our HP positive samples had the *cagA* gene. Our findings are similar to studies from Iran (65%), but less than several Indian studies (≥70%), and China (97%) [17,18]. The lower prevalence compared to other regions of India may be related to the geographic variation described in earlier studies. Although it is a limitation of this study that we did not test for the *cagA* empty site, we can assume that our isolates had intact *cagPAI* as a previous study from eastern India found that >80% of the isolates had intact *cagPAI* [19].

The EPIYA A and B motifs are common to all isolates with *cagA* and do not significantly contribute to pathogenesis, the presence of multiple C motifs or D motif have been found to be more virulent. Strains possessing the EPIYA AB with single or multiple C motifs are termed as western *cagA*, and those with EPIYA AB with D motif are termed as eastern *cagA*. As per their naming the western and eastern types are more commonly isolated from the western (Europe, Turkey, Iran, etc.) and eastern countries (China, Japan, Taiwan, etc.) and have been said to contribute to the disparity in the prevalence of gastric cancer in these regions. In our study, western *cagA* with EPIYA ABC motif was the commonest (53%) motif, and none of our samples had more than one C sequence or possessed the eastern *cagA* genotype. Our findings agree with a study from Kashmir that found that none of their isolates possessed the eastern *cagA* type, and other studies from India have found that Indian *cagA* containing HP isolates are predominantly of the western type with only a few isolates possessing the eastern *cagA* [20]. However, it should be noted that we did not include the samples with gastric cancer for analysis because very small number of patients could be recruited during the study period, which is the main limitation of the study.

VacA gene is present in all HP isolates and encodes for a potent cytotoxin that induces vacuolation and apoptosis of host cells, causes disruption of cellular signaling and immune modulation, and is associated with peptic ulcer disease. Different alleles exist for the *vacA* gene signaling, mid and intermediate region, namely *s1*, *s2*, *m1*, *m2*, *i1* and *i2*. Laboratory studies have shown that strains with the *s1*–/*m1*–type strains exhibit high toxicity, *s1*–/*m2*–type strains have low toxicity, *s2*–/*m2*–type strains are nontoxic and *s2*–/*m1*–type strains are extremely rare. In this study, *s1* allele (82.8%) was more prevalent than the *s2* allele (10.8%), a finding which is in agreement with a study by Jeyamani et al. [21]. The *s1m1* (53.8%) allele were the commonest *vacA* alleles, which is similar to a recent study from South Africa but lower than Mexico (80%) and Japan (68.2%) [22]. A significant association was found between *vacA* strains with *s2m2* and *s1/m1m2* with peptic ulcer. Although, this finding is intriguing since laboratory studies found *s2m2* strains non toxic, a study in past showed that such strains were associated with chronic gastritis

and gastric ulcer [23]. More studies with larger sample size are needed to validate this finding in our region.

The *babA2* gene encodes for the full-length *babA* protein, which helps HP colonize the stomach by binding the Lewis b, Le^b blood group antigens from the O blood group present on all gastric epithelial cells. We found that 46.2% of our samples possessed the *babA2* gene and it was not significantly associated with disease outcome. *babA2* positivity was similar to reports from Europe (44%) [24]. Western studies have found that increased expression of *babA* is associated with PUD development; the same has not been replicated in studies from Asia.

A reduced affinity to the drug due to point mutations in the domain V of the 23S ribosomal ribonucleic acid (rRNA) gene results in resistance to clarithromycin in HP. Several methods have been employed to detect these, including PCR gene amplification followed by sequencing and real-time PCR. Real-time PCR based tests usually detect the most common mutations (A2142G, A2143G), which result in clarithromycin resistance. We found a very low prevalence of resistance (11.82%) in our patient population using real-time PCR. Our findings are lower than that reported for Asia (30.9%) and even lower than the lowest rate, i.e., 18.5% reported for North America in a recent systematic review [25]. However, the findings of the above study may not be applicable to our setting as it did not include any studies from India. Our results are similar to studies from North India which reported resistance rates of 11.8% but higher than a recent study from northeast India and lower than other Indian studies which have been reviewed by Gehlot et al. [26]. A possibility of underestimating the prevalence of clarithromycin resistance exists in this study as other mechanisms leading to clarithromycin resistance were not explored.

Culture positivity of 34.4% was found in our study. It is well known that culture of HP is dependent on technical competence of the laboratory and endoscopists expertise in preparation of gastric biopsy [27]. Across the world isolation rates vary between 30% and 70% and similar rates of isolation have been reported in different studies from India [28, 29]. High rates of resistance to metronidazole (65.6%) and levofloxacin (81.2%) and lower resistance to amoxicillin (3.1%), similar observations were also reported from other studies from India [27–29]. While, this may not represent the true picture in our region because of low rate of HP isolation in this study, it is the first such study from Odisha and extensive multicentric studies are required to understand the true antibiogram of HP from Odisha.

5. Conclusion

In conclusion, this is the first study that explored the virulence factors of HP and its resistance in dyspeptic patients from this region of India. However, the findings of the study need further validation through extensive multicentric studies in this population.

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CRediT author statement

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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